

HAMILTON–BOLTZMANN COTANGENT GEOMETRY FROM DETERMINISTIC HARD-SPHERE COLLISIONS

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ABSTRACT. On an explicit perturbative regular phase chart we prove a nonautonomous biased kinetic energy estimate. The covariance is constructed first from microscopic marked cumulants; its exact dual kernel follows from the initial-plus-collision Gaussian noise representation. The Cameron–Martin inverse has domain $\text{Ran } \Sigma^{1/2}$, the action tangent is obtained by Mosco convergence, and process tightness retains the microscopic diagonal term.

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1. ROUND-TWELVE POSITIVE CLOSURE: PERTURBATIVE KINETIC CONTROL, EXACT COVARIANCE KERNEL, AND PROCESS GAUSSIANTY

The regular source chart is made quantitative. On it the biased collision operator is a small graph-bounded perturbation of the Maxwellian linearized operator. Covariance is constructed first from microscopic cumulants; its dual kernel and Cameron–Martin inverse are then identified from the local noise representation, without a hard-sphere Livšic assertion.

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1.1. A nonautonomous biased energy theorem. Let $M_{\rho,u,T}$ be the local Maxwellian determined by the prepared conserved fields. The regular chart consists of paths and collision ratios satisfying

$$cM \leq f \leq CM, \quad \|f - M\|_{W_{x,v}^{2,\infty}(M^{-1})} + \|q - 1\|_{W_w^{1,\infty}} \leq \delta, \quad 0 < q_- \leq q \leq q_+.$$

The B2 small source/time map places every regular exposed phase in this chart.

Lemma 1.1 (Perturbative biased collision gap). *For δ small, the symmetric part of the linearized biased collision operator satisfies*

$$-\langle h, L_{f,q}^{\text{sym}} h \rangle_{L^2(M^{-1})} \geq c\|(I - \Pi_5)h\|_\nu^2 - C\delta\|\Pi_5 h\|^2,$$

where Π_5 is the five-dimensional velocity-invariant projection. The constants are uniform on the chart.

Proof. At $(M, 1)$ the Dirichlet form is the classical Maxwellian form with its five-dimensional kernel. Expand $f = M + (f - M)$ and $q = 1 + (q - 1)$ in the binary collision integral. The difference is bounded, in the collision graph norm, by $C\delta\|h\|_\nu^2$. Absorb the microscopic part into the equilibrium gap; the remaining finite-rank invariant contribution gives the displayed term. $\square$

Let u solve the linearized biased balance with prepared initial data. Write $u = \Pi_5 u + (I - \Pi_5)u$; the coefficients of $\Pi_5 u$ are fields on $[0, T] \times \mathbb{T}^3$, not five scalars.

Theorem 1.2 (Variable-coefficient hypocoercive control). *For $T \leq T_*$ and the regular chart,*

$$\sup_{t \leq T} \|u_t\|_{L^2(M^{-1})}^2 + \int_0^T \|(I - \Pi_5)u_t\|_\nu^2 dt + \|\Pi_5 u\|_{L_t^2 H_x^{-1}}^2 \leq C(\|u_0\|^2 + \|F\|_{X^*}^2).$$

The transport/contact trace map is continuous in this energy space. The same estimate holds for the backward adjoint equation with endpoint data.

Proof. Multiply the microscopic equation by u/M and use Lemma 1.1. Commutators with the variable Maxwellian and q are $C\delta$ graph-bounded errors. Taking the five collision moments gives the linear acoustic system for the spatial coefficient fields, forced by the microscopic flux. Testing that system in H_x^{-1} and using the microscopic dissipation controls the macroscopic part. Endpoint and contact terms cancel through the B2 Green trace identity. Gronwall on a short interval absorbs the remaining $C(\delta + T)$ terms. Time reversal gives the adjoint estimate. $\square$

1.2. Covariance first and the exact dual kernel. For a density/contact source F , let

$$\Sigma(F, G) = \lim_{\varepsilon \rightarrow 0} \mu_\varepsilon \text{Cov}(F(\pi^\varepsilon, \Gamma^\varepsilon), G(\pi^\varepsilon, \Gamma^\varepsilon)).$$

B1 supplies the prepared initial block and B2 supplies the marked contact cumulants.

Theorem 1.3 (Joint Gaussian noise representation). *On the regular chart, Σ is the covariance of the unique linearized biased Boltzmann martingale problem*

$$d\langle u_t, \phi \rangle = \langle u_t, \mathcal{A}_{f,q}^* \phi \rangle dt + dM_t(\phi),$$

with prepared initial covariance and bracket

$$d\langle M(\phi), M(\psi) \rangle_t = \int q \Delta \phi \Delta \psi dA_f dt.$$

For a dual source pair, variance is zero exactly when its backward adjoint solution has zero projected initial source and zero local collision-noise coefficient. This kernel is the closed balance/endpoint gauge; no long-time Livšic theorem is invoked.

Proof. Differentiate the B2 marked Hamiltonian twice and use the B1 Schur complement at time zero. The result is the displayed linear Gaussian martingale problem. Solve its backward adjoint equation by Theorem 1.2. It represents every source as the sum of an initial Gaussian pairing and a stochastic integral against the collision noise. Orthogonality of those two terms makes the variance zero precisely when both coefficients vanish. Reversing the weak balance identifies this condition with the endpoint-normalized representation gauge. $\square$

Proposition 1.4 (Typed cotangent and Cameron–Martin spaces). *The continuous dual source space is quotiented by $\ker \Sigma$, a closed linear subspace. The primal Cameron–Martin form is*

$$\mathfrak{q}(z) = \begin{cases} \|\Sigma^{-1/2} z\|^2, & z \in \text{Ran } \Sigma^{1/2}, \\ +\infty, & \text{otherwise.} \end{cases}$$

It has no nonzero null vector. The dual quotient and the primal form domain are not identified with one another.

Proof. The covariance kernel is closed by continuity. The spectral theorem for the positive covariance operator gives the displayed closed extended quadratic form and its exact domain. Positivity is strict on that domain after the dual kernel has been quotiented. $\square$

1.3. Second-order action and Mosco convergence.

Theorem 1.5 (Second epi-derivative). *Let I be the B2 prepared path action. At a regular exposed path, its second epi-derivative on the balanced Gelfand triple equals $\mathfrak{q}$. Finite conditional-expectation projections of the Gaussian noise give an increasing sequence of closed forms converging to $\mathfrak{q}$ in the Mosco sense.*

Proof. On a finite family of tests, analytic pressure duality gives the inverse covariance Hessian. Realize all projections on the same initial-plus-collision Gaussian noise space; conditional expectation onto increasing noise sigma-fields makes the energies monotone. The martingale convergence theorem

gives the liminf and recovery of the closed forms. The energy estimate of Theorem 1.2 identifies the closure of smooth balanced variations with this Gaussian control space. Hence the second epi-derivative is exactly $\mathbf{q}$. $\square$

1.4. Process tightness with the microscopic diagonal term. Let Z_t^ε be the centered $\sqrt{\mu_\varepsilon}$ density/contact field.

Lemma 1.6 (Aldous increment estimate). *For every bounded stopping time τ and $0 \leq h \leq 1$,*

$$\mathbb{E}|Z_{\tau+h}^\varepsilon - Z_\tau^\varepsilon|^4 \leq C \left(h^2 + \frac{h}{\mu_\varepsilon} \right).$$

The maximal jump in the declared distribution norm converges to zero in probability.

Proof. The fourth cumulant of a contact increment contains one connected diagonal term of size h/μ_ε and pairings of size h^2 . The localized B2 cluster density bounds all remaining connected diagrams by the same right side. Each microscopic density/contact jump has size $O(\mu_\varepsilon^{-1/2})$ in the negative test norm; the exponential contact count bound controls their maximum. $\square$

Theorem 1.7 (Prepared joint process CLT). *The fields Z^ε converge in the weighted distribution Skorokhod space to the Gaussian martingale problem of Theorem 1.3. All density/contact cross-brackets and the prepared initial covariance are included.*

Proof. Finite-dimensional cumulants converge by normal pressure convergence and all orders at least three vanish after central scaling. Lemma 1.6 gives Aldous tightness; Mitoma's criterion lifts it to the distribution space. Closedness of the B2 balance graph identifies every limit, and uniqueness of the linear Gaussian martingale problem gives full convergence. $\square$

Theorem 1.8 (Round-twelve B3 closure). *On the explicit perturbative regular chart, the prepared hard-sphere tangent has a proved nonautonomous energy estimate, exact covariance kernel, correctly typed Cameron–Martin form, second-order action, and joint process Gaussian limit.*

Proof. Combine Theorems 1.2, 1.3, 1.5, and 1.7, with Proposition 1.4. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states positive theorem closure in the declared model-specific regular regime. Cross-paper inputs follow the round-twelve dependency ledger. Repository source, proof-structure, counterexample, and build gates are internal reproducibility checks and do not replace independent external mathematical review.

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